

# Chapter 5: States of Matter & Stoichiometry

*Gas Laws, Phase Diagrams, Mole Concept & Reaction Stoichiometry*

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## 5.1 Gas Laws & Kinetic Molecular Theory

Gases exhibit predictable macroscopic behavior described by the Ideal Gas Law:

$$PV = nRT$$

Where  $P$  is pressure,  $V$  is volume,  $n$  is moles,  $R$  is the ideal gas constant, and  $T$  is temperature in Kelvin. Dalton's Law of Partial Pressures states that total pressure equals the sum of individual partial pressures.

## 5.2 The Mole Concept & Stoichiometry

One mole contains Avogadro's number ( $N_A \approx 6.022 \times 10^{23}$ ) representative particles.

Stoichiometry uses balanced chemical equations to calculate reactant and product masses, limiting reagents, and theoretical percentage yields.

## 5.3 Solutions & Concentration Metrics

Solutions are homogeneous mixtures. Concentrations are expressed in multiple formats:

- **Molarity (M):** Moles of solute per liter of solution ( $\text{mol/L}$ ).
- **Molality (m):** Moles of solute per kilogram of solvent ( $\text{mol/kg}$ ).
- **Mole Fraction ( $X_i$ ):** Ratio of moles of component  $i$  to total moles.